What is Machine Learning?

A Formal Definition

- **Machine Learning** is formally defined as "the field of study that gives computers the **ability to learn without being explicitly programmed**."
- This definition is attributed to **Arthur Samuel**, a pioneer in the field of AI and computer gaming.

Core Concepts 👨‍💻

- **Learning from Experience:** The core idea is that a program can improve its performance on a task by learning from data or experience.
- **Arthur Samuel's Checkers Program (1950s):**
  - Samuel, who was not a great checkers player himself, wrote a program that learned to play checkers.
  - He had the computer play **tens of thousands of games against itself**.
  - By observing which board positions led to wins versus losses, the program learned to identify good and bad positions, eventually becoming a better player than its creator.
  - This illustrates a key principle: in general, **the more data (experience) a learning algorithm has, the better it will perform**.

Major Types of Machine Learning

The two main categories of machine learning algorithms are:

1. **Supervised Learning:** This is the most widely used type of machine learning in real-world applications and is the focus of the first two courses in this specialization.
2. **Unsupervised Learning:** This will be covered in the third course.

Focus on Practical Application 🛠️

- Beyond just learning the algorithms ("the tools"), this course emphasizes the **practical skills needed to apply them effectively**.
- Knowing *how* and *when* to use different techniques is crucial for building successful machine learning systems and avoiding common pitfalls that can waste significant time and effort.

Supervised Learning

Supervised learning is the most common type of machine learning and is responsible for most of its economic value today. It's all about learning a mapping from an **input (x)** to an **output (y)** based on labeled examples.

How it Works 🧠

The core idea is to "supervise" the algorithm by giving it a dataset that includes the "right answers."

1. **Training:** The algorithm is fed a large amount of training data, where each data point consists of an input `x` and its corresponding correct output label `y`.
2. **Learning:** By seeing many `(x, y)` pairs, the algorithm learns the underlying relationship between the inputs and outputs.
3. **Prediction:** Once trained, the model can be given a **new input `x`** that it has never seen before and produce a prediction, or a good guess, for the output `y`.

Examples of Supervised Learning

Application	Input (x)	Output (y)
Spam filtering	An email	Spam (1) or Not Spam (0)
Speech recognition	An audio clip	Text transcript
Machine translation	A sentence in English	The sentence in Spanish
Online advertising	Ad info + user info	Will the user click? (Yes/No)
Self-driving cars	Image from a camera	Position of other cars
Visual inspection	Picture of a phone	Defect (1) or No Defect (0)

Types of Supervised Learning: Regression

One of the two major types of supervised learning is called **regression**.

- **Goal:** To predict a **continuous numerical value**. This means the output can be any number within a range.
- **Example: Housing Price Prediction**

- You provide the model with a dataset of houses, where the input `x` is the **size of the house** (e.g., in square feet) and the output `y` is its **selling price**.
- The algorithm learns the relationship by fitting a line or a curve to this data.
- It can then predict a price for a house size it has never seen before.

Types of Supervised Learning: Classification

Classification is the second major type of supervised learning.

- **Goal:** To predict a **discrete category** (or class) from a small, finite set of possible outputs.
- **Key Difference from Regression:**
  - **Regression** predicts a continuous number from an infinite range (e.g., a house price).
  - **Classification** predicts one of a few distinct outcomes (e.g., is this email spam or not?).

Example: Breast Cancer Detection 🩺

- **Input (x):** Features of a tumor, such as its **size**.
- **Output (y):** A category — either **Malignant** (cancerous) or **Benign** (not cancerous).
- **Data Representation:** These categories are often mapped to numbers for the model to process, for example:
  - Benign = **0**
  - Malignant = **1**

Multi-Class Classification

- Classification is not limited to just two categories. A problem can have multiple possible output classes.
- For example, a model could be trained to distinguish between "Benign," "Cancer Type 1," and "Cancer Type 2."
- In this case, the output `y` could be 0, 1, or 2, each representing one of the distinct classes.

Using Multiple Features & The Decision Boundary

- Models can use multiple input features to make more accurate predictions.
- **Example:** Using both a patient's **age** and the **tumor size**.
- When using multiple features, the learning algorithm's goal is to find a **decision boundary**—a line or curve that best separates the different classes in the data.
- For a new patient, the model determines which side of this boundary their data point falls on to make its classification.

Summary: Regression vs. Classification

Feature	Regression	Classification
Output Type	Continuous (a number)	Discrete (a category/class)
Goal	Predict a quantity	Predict a label
Example	Predicting house prices	Identifying spam emails

Unsupervised Learning

Unsupervised learning is the second major type of machine learning. It is used when the data provided to the algorithm has **no output labels** (`y`).

How it Works 🧠

- Unlike supervised learning, where the goal is to map an input `x` to a known output `y`, unsupervised learning is given a dataset consisting only of inputs.
- The algorithm's task is to find **structure, patterns, or something interesting** within the data on its own, without any "right answers" to guide it.
- **Example:** Given data on patients' tumor sizes and ages, but *without* labels indicating if the tumors are benign or malignant, an unsupervised algorithm might identify that the data points naturally fall into two distinct groups.

Clustering: A Key Type of Unsupervised Learning

One of the most common types of unsupervised learning is **clustering**.

- **Goal:** A clustering algorithm takes unlabeled data and automatically groups it into different clusters based on similarity.

Examples of Clustering in Action

1. **Google News:**
    - Every day, the algorithm analyzes hundreds of thousands of news articles.
    - It identifies articles with similar words and topics (e.g., "panda," "twins," "zoo") and groups them together into a single news story.
    - It does this **without human supervision**, automatically discovering the day's emerging topics.
2. **Genetics (DNA Analysis):**
    - Given genetic data from many individuals, a clustering algorithm can group people into different categories based on their gene activity patterns.
    - This can help researchers discover different genetic profiles or types of individuals within a population without knowing those types in advance.
3. **Market Segmentation:**
    - Companies use clustering to analyze their customer databases.
    - The algorithm groups customers into different market segments based on their behaviors, preferences, or motivations.

- This allows the company to better understand and serve different types of customers. For example, an education company might find distinct clusters of learners motivated by career growth vs. pure knowledge acquisition.

Unsupervised Learning: Formal Definition & Other Types

Formal Definition

- **Supervised Learning:** The data provided is a set of **(x, y)** pairs, where x is the input and y is the correct output label.
- **Unsupervised Learning:** The data provided is only a set of inputs **(x)**, with **no corresponding output labels**. The goal is for the algorithm to discover interesting structures or patterns within this unlabeled data.

Types of Unsupervised Learning

Beyond clustering, there are other important types of unsupervised learning algorithms.

1. **Clustering (Recap):**
   - Groups similar data points together.
   - *Example:* Grouping news articles or segmenting customers.
2. **Anomaly Detection:**
   - Used to identify rare or unusual data points/events.
   - *Example:* **Fraud detection**. An algorithm can analyze many financial transactions to learn what a "normal" transaction looks like and then flag unusual ones that might be fraudulent.
3. **Dimensionality Reduction:**
   - Compresses a large dataset into a much smaller one while losing as little information as possible.
   - This is useful for data visualization, saving storage space, and speeding up other learning algorithms.

Reinforcing the Concepts (Quiz Summary)

Here's how to categorize different problems:

- **Spam Filtering (Supervised):** Requires a dataset of emails already labeled as "spam" or "not spam."
- **Grouping News Articles (Unsupervised):** Uses a clustering algorithm to find related articles without any pre-existing labels.
- **Market Segmentation (Unsupervised):** Uses a clustering algorithm to discover customer groups automatically from unlabeled data.
- **Diabetes Diagnosis (Supervised):** Requires a dataset of patient records labeled as

"diabetes" or "no diabetes," similar to the breast cancer classification example.

Jupyter Notebooks: The ML Workspace 📓

The **Jupyter Notebook** is the most widely used tool and development environment for machine learning and data science practitioners. This course uses the same professional-grade Jupyter Notebook environment that developers use in major tech companies.

Types of Labs in this Course

1. **Optional Labs:**
    - These labs are designed to help you understand how machine learning code works without requiring you to write any yourself.
    - You only need to read the provided text and **run the existing code cells**.
    - They are completely optional and are not graded. Their purpose is to give you a deeper feel for the concepts.
2. **Practice Labs:**
    - Starting in Week 2, these labs will give you the opportunity to **write your own code**.

How to Use a Jupyter Notebook

A notebook is made up of blocks called **cells**. There are two main types:

- **Markdown Cells:** These contain formatted text, like headings, descriptions, and explanations.
- **Code Cells:** These contain executable code (in this course, Python).

Running a Cell:

- To run the code or render the text in any selected cell, press **Shift + Enter** on your keyboard.
- You are encouraged to experiment by editing the code in the optional labs and running it again to see what happens.